

Zephyr Workspace Setup Guide

Setup steps for the latest Zephyr inside myworkspace/ using Python 3.10.

Note: Python 3.9 (Anaconda) is too old — Zephyr's scripts require `patool>=2.0.0` which needs Python ≥ 3.10 . Use the system `python3.10` instead.

Step 1 — Create a virtual environment with west

```
cd /home/jovinn/workspac/My-Zephyr-project/myworkspace
python3.10 -m venv .venv
source .venv/bin/activate
pip install west
```

Step 2 — Initialize the Zephyr workspace

```
unset ZEPHYR_BASE # clear any existing ZEPHYR_BASE before init
west init .
west update
```

Downloads Zephyr kernel + all HAL/module dependencies (~1–2 GB, takes a few minutes).

Step 3 — Install Zephyr Python requirements

```
pip install -r zephyr/scripts/requirements.txt
```

Step 4 — Install the Zephyr SDK (ARM toolchain for nRF52840)

Check if the SDK is already installed first:

```
ls ~/zephyr-sdk-* 2>/dev/null && echo "SDK found" || echo "SDK not installed"
```

If not installed:

```
cd ~
wget https://github.com/zephyrproject-rtos/sdk-ng/releases/download/v0.16.0/zephyr-sdk-0.16.0_linux-x86_64.tar.xz
tar xf zephyr-sdk-0.16.0_linux-x86_64.tar.xz
```

```
cd zephyr-sdk-0.16.8
./setup.sh
```

Step 5 — Build a project

Run these commands at the start of every new terminal session:

```
source /home/jovin/workspace/My-Zephyr-project/myworkspace/.venv/bin/activ
export ZEPHYR_BASE=/home/jovin/workspace/My-Zephyr-project/myworkspace/zep
```

Then navigate to any project and build:

```
cd /home/jovin/workspace/My-Zephyr-project/src/2026-05-16_Introduction_to_
west build -b nrf52840dk/nrf52840
```

Flash to the board:

```
west flash
```

Quick reference — session setup (copy-paste)

```
source /home/jovin/workspace/My-Zephyr-project/myworkspace/.venv/bin/activ
export ZEPHYR_BASE=/home/jovin/workspace/My-Zephyr-project/myworkspace/zep
```

Notes

- Use **python3.10**, not Anaconda's python3.9 — Zephyr requires patool $\geq$ 2.0.0 which needs Python $\geq$ 3.10.
- ZEPHYR_BASE must **not** be set before running `west init` — unset it first if it points to `~/ncs/v2.4.2`.
- The `.venv` folder is inside `myworkspace/` which is excluded from git (`.gitignore`).
- If `west` is not found after activating the venv, re-run `pip install west` inside the activated environment.
- SDK version 0.16.8 supports Zephyr v3.4 and later.